

MCP Data Analysis Agent

Compact operational guide - current contract

Local, governed database access for MCP clients. The server is credential-free globally; each data project opts in through its own private source file.

What it does

- Runs as a stdio MCP server for Codex, Claude Code, Copilot, Cline, Cursor, Windsurf, and Continue.
- Uses one project-local **.mcp-data-source** file for an SQLite path/URL or PostgreSQL URL.
- Performs bounded read-only probes, validates SQL, limits execution, and stores auditable evidence in **observability/**.
- Offers schema discovery, governed queries, metrics, recipes, comparisons, reports, and an explicit deterministic retail demo.

It never uploads data, hosts a network service, creates database users, stores credentials in client configuration, or changes a source database.

Tell an agent to install it

Use this instruction with an agentic coding client:

Please install MCP Data Analysis Agent from this repository using its documented install.sh. Configure only credential-free global MCP client entries. Do not create a database, inspect or modify my project, or configure a source until I explicitly choose a demo or provide a source URL.

Public repository install

```
curl -fsSL https://raw.githubusercontent.com/hassanvfx/mcp-data-analysis-agent/main/install.sh | bash
```

Already cloned checkout

```
./install.sh --local
```

Both routes add only static **mcp-data-mcp --source-file .mcp-data-source** entries. Restart or trust the MCP client if it asks.

Operating model

Scope	Contains	Never contains
Global	Executable and static client entry	URL, password, project setup
Data project	Source file, optional policy, evidence	Repository clone, Git, Typst, PostgreSQL CLI

```
install -> preflight -> demo or real source -> ready probe -> inspect schema -> analyze -> optional cleanup
```

Configure a data project

Start with the demo

```
mcp-data-cli preflight mcp-data-cli demo start --yes mcp-data-cli preflight mcp-data-cli schema data mcp-data-cli query data 'SELECT id, name, stock FROM products ORDER BY id' --limit 10
```

The confirmed demo creates only **.mcp-data/playground.sqlite** and a mode-0600 **.mcp-data-source**. Git ignore entries are added only when safe.

Configure a real source

```
mcp-data-cli configure-source /absolute/path/to/analytics.sqlite --yes mcp-data-cli preflight # PostgreSQL:
mcp-data-cli configure-source 'postgresql://readonly_user:password@db.example:5432/analytics' --yes
```

The source file is the sole runtime secret. It must be a regular non-symlink file with exactly one absolute SQLite path/URL or PostgreSQL URL. Never commit it.

Preflight	Meaning and action
source_configuration_required	No source file. Start the demo or configure a real source.
source_configuration_invalid	Unsafe or malformed source/policy. Correct local configuration.
source_unavailable	Read-only probe failed. Check reachability and account privileges.
ready	Read-only SELECT 1 succeeded. Discover schema and analyze.

Analyze and retain evidence

```
mcp-data-cli schema data mcp-data-cli sql data 'SELECT id, name FROM products WHERE id = :id' --params '{"id": 1}' mcp-data-cli explain data 'SELECT id, name FROM products WHERE id = :id' --params '{"id": 1}' mcp-data-cli query data 'SELECT id, name FROM products ORDER BY id' --limit 25
```

```
mcp-data-cli task-begin 'Inventory review' 'Find low-stock products.' mcp-data-cli observe <task-id> mcp-data-cli task-complete <task-id> 'Findings recorded.'
```

Receipts, lifecycle events, and evidence remain project-local under **observability/**. HTML and CSV reports need no renderer; Parquet needs its package extra, and PDF needs optional Typst.

Optional governance

```
mcp-data-cli configure-policy --yes
```

This creates non-secret policy/catalog/recipe starters only when none already exist. It never infers policy, overwrites governance, or creates a source file.

Client fallback and cleanup

Global entries resolve the source file relative to the MCP working directory. Open the target project and call MCP **preflight**. If a client does not preserve that directory, configure a project entry:

```
mcp-data-cli setup --client <client> --apply
```

That entry uses a static absolute **--project-root** and still contains no URL or credential.

Preview and apply cleanup

```
mcp-data-cli uninstall --clients --demo mcp-data-cli uninstall --clients --demo --apply --yes uv tool uninstall  
mcp-data-analysis-agent
```

Cleanup removes only exact managed client entries and managed demo artifacts. It preserves custom sources, governance files, observability evidence, and unrelated client settings. A custom source that replaced the demo is retained.

Safety contract

- Only parameterized read-only SELECT or WITH queries are permitted.
- Mutations, DDL, attachments, multi-statements, unsafe functions, restricted fields, and unsafe paths are denied.
- SQLite and PostgreSQL sessions use read-only controls; PostgreSQL still requires a least-privilege database account.
- Source URLs and secret-like values are redacted from output and evidence.

Current source: README.md | Operations: docs/OPERATIONS.md | Security: docs/SECURITY.md